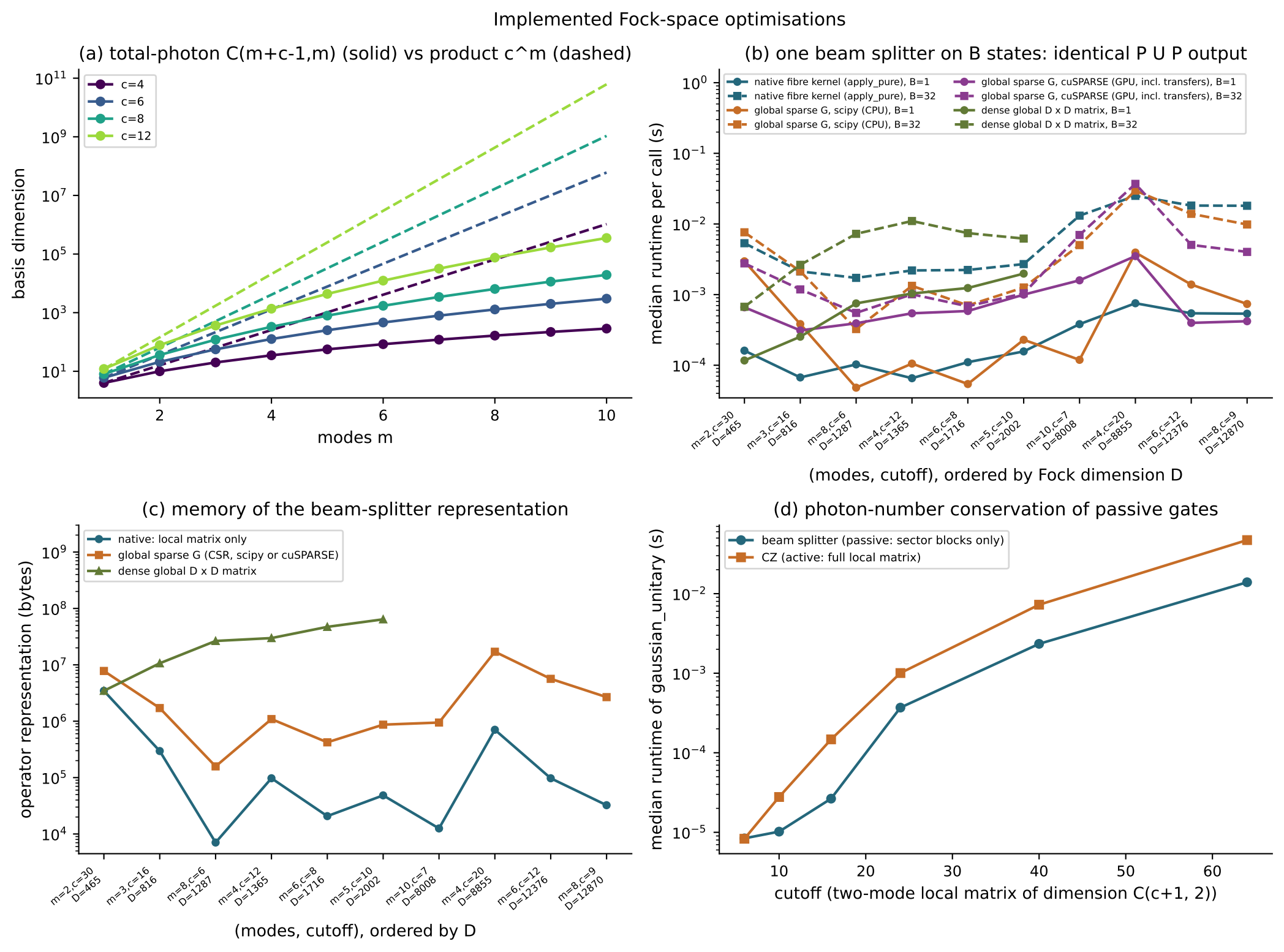
Implemented Fock-space optimisations
(a) total-photon C(m+c-1,m) (solid) vs product c^m (dashed)
basis dimension
modes m
c=4
c=6
c=8
c=12
(b) one beam splitter on B states: identical P U P output
median runtime per call (s)
native fibre kernel (apply_pure), B=1
native fibre kernel (apply_pure), B=32
global sparse G, scipy (CPU), B=1
global sparse G, scipy (CPU), B=32
global sparse G, cuSPARSE (GPU, incl. transfers), B=1
global sparse G, cuSPARSE (GPU, incl. transfers), B=32
dense global D x D matrix, B=1
dense global D x D matrix, B=32
m=2,c=30 D=465
m=3,c=16 D=816
m=8,c=6 D=1287
m=4,c=12 D=1365
m=6,c=8 D=1716
m=5,c=10 D=2002
m=10,c=7 D=8008
m=4,c=20 D=8855
m=6,c=12 D=12376
m=8,c=9 D=12870
(modes, cutoff), ordered by Fock dimension D
(c) memory of the beam-splitter representation
operator representation (bytes)
native: local matrix only
global sparse G (CSR, scipy or cuSPARSE)
dense global D x D matrix
(modes, cutoff), ordered by D
(d) photon-number conservation of passive gates
median runtime of gaussian_unitary (s)
beam splitter (passive: sector blocks only)
CZ (active: full local matrix)
cutoff (two-mode local matrix of dimension C(c+1, 2))